

SARAB, Iran, Islamic Republic of

WMO: 407100

Lat: 37.933N

Lon: 47.533E

Elev: 5518

StdP: 11.99

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	2.6	-6.2	4.8	-0.3	-1.4	6.2	5.4	29.5	33.2	25.7	34.7	1.0	30	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	29.4	90.3	59.1	87.5	58.5	84.6	58.1	62.8	80.7	61.6	79.4	60.4	78.2	8.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.7	84.2	69.0	55.3	80.1	68.1	54.0	76.2	67.2	31.3	80.6	30.2	79.2	29.4	78.1	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.7	22.0	18.8	DB	-8.3	95.3	5.4	2.0	-12.2	96.7	-15.3	97.9	-18.3	99.0	-22.2	100.4	(4)
(5)			WB	-8.6	65.8	5.2	2.1	-12.4	67.3	-15.5	68.5	-18.4	69.7	-22.2	71.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	24.6	29.7	39.0	47.5	56.7	64.8	69.6	69.7	61.6	50.8	38.3	29.2		
	DBStd	17.01	9.90	8.65	7.94	6.54	4.78	4.48	4.82	4.65	5.13	5.76	7.32	9.43		
	HDD50	2898	789	570	352	120	6	0	0	0	1	61	353	646		
	HDD65	6392	1254	990	806	526	260	56	12	13	123	441	800	1111		
	CDD50	2367	0	0	11	43	213	445	607	611	349	85	3	0		
	CDD65	387	0	0	0	0	2	51	154	159	21	0	0	0		
	CDH74	5581	0	0	0	14	78	892	1802	2090	683	22	0	0		
	CDH80	1906	0	0	0	0	3	229	681	835	158	0	0	0		
(14) Wind	WSAvg	7.0	5.8	6.4	8.5	7.8	7.1	8.1	8.6	7.9	6.7	6.1	5.3	5.4		
(15) Precipitation	PrecAvg	12.3	0.8	0.9	1.3	2.2	2.3	0.7	0.5	0.3	0.4	1.0	1.2	0.8		
	PrecMax	15.9	1.7	2.0	2.4	3.9	4.0	2.0	1.6	0.9	1.3	3.2	3.8	1.9		
	PrecMin	7.5	0.3	0.2	0.4	0.7	0.4	0.0	0.0	0.0	0.0	0.0	0.2	0.1		
	PrecStd	2.2	0.3	0.4	0.5	0.9	0.9	0.6	0.4	0.3	0.4	0.8	0.8	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	52.0	67.5	75.8	80.0	89.8	94.1	93.5	87.7	76.8	62.2	54.5		
		MCWB	37.6	39.5	44.3	51.8	54.6	59.0	60.1	58.9	56.4	51.3	45.0	40.5		
	2%	DB	43.4	48.1	62.0	69.1	76.0	85.2	90.3	90.6	84.3	73.0	58.0	49.1		
		MCWB	35.4	37.2	43.4	49.1	52.8	57.4	60.3	58.4	55.4	50.3	43.2	39.0		
	5%	DB	40.7	44.8	57.4	64.8	73.2	82.4	87.0	88.0	81.5	69.5	54.5	45.5		
		MCWB	33.8	36.0	41.5	47.8	52.3	56.8	59.6	58.2	54.8	49.1	42.6	37.3		
	10%	DB	38.0	41.9	53.3	61.0	70.2	79.1	83.6	85.3	78.1	65.7	51.1	42.3		
		MCWB	32.4	34.8	40.1	46.1	51.7	56.0	59.6	58.1	54.1	48.3	41.5	35.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.8	41.1	46.6	53.3	57.3	62.3	64.8	64.5	60.5	54.5	48.1	42.6		
		MCDB	45.6	49.2	60.5	67.5	73.6	82.6	83.8	80.8	76.8	68.8	57.1	50.8		
	2%	WB	36.9	38.8	44.3	50.8	55.5	60.4	63.3	62.5	58.7	52.2	46.2	40.3		
		MCDB	41.6	45.5	59.5	64.5	71.1	79.3	81.9	79.2	75.1	66.9	53.1	47.1		
	5%	WB	34.5	36.9	42.6	49.2	54.1	59.0	62.0	61.2	57.3	50.9	44.3	38.1		
		MCDB	39.3	43.1	55.6	61.7	68.5	77.5	80.1	78.8	74.0	64.7	51.2	44.0		
	10%	WB	32.5	35.2	40.9	47.6	52.8	57.6	60.9	60.2	56.0	49.6	42.3	36.0		
		MCDB	37.3	41.2	51.4	58.8	66.5	75.2	78.8	78.0	72.4	62.4	49.9	41.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	18.3	20.6	22.0	23.6	27.4	26.9	29.4	30.8	26.3	21.1	19.0		
		MCDBR	19.6	22.2	28.3	29.1	28.2	31.7	31.6	33.8	35.8	32.9	27.9	23.9		
		MCWBR	12.7	13.0	14.1	13.0	10.4	11.1	10.0	10.9	13.3	14.7	15.5	14.9		
	5% WB	MCDBR	15.9	18.1	24.0	24.7	25.3	28.5	27.2	28.8	31.1	28.2	20.5	20.1		
		MCWBR	10.8	11.0	12.5	11.9	10.3	11.1	10.2	10.5	12.6	13.5	11.8	13.0		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.351	0.414	0.494	0.488	0.467	0.509	0.467	0.409	0.402	0.335	0.305		
	taud		2.276	2.151	1.997	1.835	1.891	1.971	1.885	1.983	2.119	2.084	2.257	2.314		
	Ebn at Noon		272	273	266	251	255	260	248	255	264	251	260	266		
	Edh at Noon		31	40	52	65	62	57	62	55	45	42	31	28		
(44) All-Sky Solar Radiation	RadAvg		776	1054	1309	1594	2014	2439	2385	2180	1769	1216	816	666		
	RadStd		71	122	132	155	110	134	91	87	97	103	105	85		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (10 neighbors)	+0.59	N/A	N/A	+1.75	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page